

Reflect Points Across Axes

Lesson 9-7

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

$(3, 2)$ reflected over the y -axis becomes $(-3, 2)$ — x changes sign, y stays

Reflection

A flipped image of a point on the other side of a line, the same distance away.

$(4, 3) \rightarrow (4, -3)$: the y flips from $+3$ to -3 , like folding the paper along the horizontal line

x-axis

The line that goes across the grid. Flipping over it changes the y sign.

$(4, 3) \rightarrow (-4, 3)$: the x flips from $+4$ to -4 , like folding the paper along the vertical line

y-axis

The line that goes up the grid. Flipping over it changes the x sign.

A butterfly's wings are symmetric — fold it down the middle and both sides match

Symmetry

When a shape looks the same on both sides of a line.

..., -3, -2, -1, 0, 1, 2, 3, ...

Integer

Whole numbers and their opposites, like -2 , -1 , 0 , 1 , 2 .

Key Ideas & Notes

- Captain Vega discovers that her treasure map has a magic mirror trick!
- When she folds the map along the x-axis, some clues match up perfectly with clues on the other side.
- For example, a marker at $(2, 4)$ has a matching marker at $(2, -4)$ when reflected over the x-axis.
- Another pair reflects over the y-axis: $(5, 3)$ and $(-5, 3)$.
- Plot each original point and then plot its reflection across the given axis.

Think About It

- When $(2, 4)$ reflects over the x-axis to $(2, -4)$, which coordinate changed?
- When $(5, 3)$ reflects over the y-axis to $(-5, 3)$, which coordinate changed?
- What stays the same in each reflection?

My Notes

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

When you reflect the point (3, 2) over the y-axis, which coordinate changes its sign and which stays the same? Why?

Level 1 support

Start here: Reflecting over the y-axis changes the ____ coordinate because ____.

The ____ coordinate changes sign.

La coordenada ____ cambia de signo.

The ____ coordinate stays the same.

La coordenada ____ se queda igual.

Word bank: reflection x-axis y-axis ordered pair coordinate plane

Listen for: Student states reflecting over the y-axis flips the sign of the x-coordinate while the y-coordinate stays the same, so (3,2) becomes (-3,2).

Level 2

Push students to explain why the reflected point is the same distance from the y-axis as the original.

- The reflected point is the same distance from the axis because ____.
- A reflection is like a ____ over the line.

EXPLORE

Reflect $(4, -5)$ over the x-axis. Talk through how the coordinates change and where the new point lands.

Level 1 support

Start here: Reflecting over the x-axis changes the ____ coordinate, so the point becomes ____.

Over the x-axis, the ____ coordinate flips sign.

Sobre el eje x, la coordenada ____ cambia de signo.

So $(4, -5)$ becomes ____.

Entonces $(4, -5)$ se convierte en ____.

Word bank: reflection x-axis y-axis ordered pair coordinate plane

Listen for: Student flips the sign of the y-coordinate, so $(4, -5)$ becomes $(4, 5)$, and explains the x stays the same.

Level 2

Ask students to predict where $(4, -5)$ lands if reflected over the y-axis instead, and compare the two results.

- Reflected over the y-axis, $(4, -5)$ becomes ____.
- The two reflections are different because ____.

CONNECT

How is reflecting a point over an axis like looking at it in a mirror placed on that axis?

Level 1 support

Start here: A reflection is like a mirror because ____.

The axis acts like the ____.

El eje actua como el ____.

The image is the same distance away but on the ____ side.

La imagen esta a la misma distancia pero del ____ lado.

Word bank: reflection x-axis y-axis coordinate plane ordered pair

Listen for: Student connects the axis to a mirror line and the reflected point to a mirror image, equal distance on the opposite side.

Level 2

Push students to generalize a rule: what changes when you reflect over the x-axis versus the y-axis?

- Reflecting over the x-axis changes ____, while reflecting over the y-axis changes ____.
- A rule I can always use is ____.

REFLECT

A classmate reflected $(2, 3)$ over the x -axis and got $(-2, 3)$. What did they do wrong, and what is the correct image?

Level 1 support

Start here: The mistake was changing the ____ coordinate, but the correct image is ____.

Over the x -axis, they should have changed the ____ coordinate.

Sobre el eje x , debieron cambiar la coordenada ____.

The correct image is ____.

La imagen correcta es ____.

Word bank: reflection x-axis y-axis ordered pair coordinate plane

Listen for: Student identifies they flipped x instead of y ; the correct image over the x -axis is $(2, -3)$.

Level 2

Ask students to critique how to remember which coordinate changes for each axis.

- To remember which coordinate changes, I think ____.
- Reflecting over the x -axis affects the ____ value because ____.

Guided Examples

Example 1

What is the reflection of $(4, -3)$ over the x-axis?

Solution: Reflecting over the x-axis changes the sign of the y-coordinate. The x stays the same (4), and the y changes from -3 to 3. The answer is $(4, 3)$.

Answer: B. $(4, 3)$

Example 2

What is the reflection of $(-2, 6)$ over the y-axis?

Solution: Reflecting over the y-axis changes the sign of the x-coordinate. x goes from -2 to 2, and y stays 6. The answer is $(2, 6)$.

Answer: A. $(2, 6)$

Example 3

Which point is on the y-axis?

Solution: Points on the y-axis have x-coordinate = 0. $(0, 5)$ is on the y-axis.

Answer: A. $(0, 5)$

Write About the Math

The Writing Revolution

I can explain my work using the words reflection, x-axis, y-axis, and symmetry.

1. Kernel Sentence subject + verb

Model: Reflection is a flipped image of a point on the other side of a line, the same distance away.
Reflexión es una imagen volteada de un punto al otro lado de una línea, a la misma distancia.

Write a kernel sentence about reflection. Use a subject and a verb.

Escribe una oración base sobre reflexión. Usa un sujeto y un verbo.

2. Sentence Expansion because • but • so

Kernel: Reflection matters in math
Reflexión importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Reflection matters in math because ____.
Reflexión importa en matemáticas porque ____.

but
pero

Reflection matters in math, but ____.
Reflexión importa en matemáticas, pero ____.

so
entonces

Reflection matters in math, so ____.
Reflexión importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement *Afirmación*

Tell one true fact about reflection.
Di un hecho verdadero sobre reflection.

Reflection ____.

Question *Pregunta*

Ask a question about reflection.
Haz una pregunta sobre reflection.

How does ____ ?

¿Cómo ____ ?

Exclamation *Exclamación*

Show excitement about reflection.
Muestra entusiasmo sobre reflection.

Wow, ____ !

¡Guau, ____ !

Command *Mandato*

Tell a partner what to do with reflection.
Dile a un compañero qué hacer con reflection.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

The ____ **coordinate changes sign.**

La coordenada ____ *cambia de signo.*

The ____ **coordinate stays the same.**

La coordenada ____ *se queda igual.*

Over the x-axis, the ____ **coordinate flips sign.**

Sobre el eje x, la coordenada ____ *cambia de signo.*

Try It

Solve on your own. Check the answer key when you are done.

1. Which point is on the y-axis?

- A. (0, 5)
- B. (5, 0)
- C. (3, 3)
- D. (-2, 4)

Show your work:

2. Point A is at (-3, 5). It is reflected over the x-axis to A', then A' is reflected over the y-axis to A''. What are the coordinates of A''?

- A. (3, -5)
- B. (-3, -5)
- C. (3, 5)
- D. (-3, 5)

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A triangle has vertices at $A(1, 2)$, $B(4, 2)$, and $C(4, 6)$. If you reflect the entire triangle over the y -axis, what are the new vertices? What do you notice about the size and shape of the reflected triangle compared to the original?

Sentence starter: The reflected vertices are $A'(\underline{\hspace{1cm}})$, $B'(\underline{\hspace{1cm}})$, $C'(\underline{\hspace{1cm}})$. I found each by $\underline{\hspace{1cm}}$. The reflected triangle is $\underline{\hspace{1cm}}$ compared to the original because reflections $\underline{\hspace{1cm}}$.

Show your work:

Reflect — Exit Ticket

What is the reflection of $(-5, 2)$ over the y -axis?

- A. $(-5, -2)$
- B. $(5, -2)$
- C. $(5, 2)$
- D. $(2, -5)$

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. $(0, 5)$ — *Points on the y -axis have x -coordinate = 0. $(0, 5)$ is on the y -axis.*
2. **Try It 2:** A. $(3, -5)$ — *Over x -axis: $(-3, 5) \rightarrow (-3, -5)$. Then over y -axis: $(-3, -5) \rightarrow (3, -5)$. Both coordinates changed sign.*
3. **Exit Ticket:** C. $(5, 2)$ — *Reflecting over the y -axis changes the sign of the x -coordinate. x goes from -5 to 5 , and y stays 2 . The reflection is $(5, 2)$.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Reflection is a flipped image of a point on the other side of a line, the same distance away.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).